



General histology

***Hawler Medical University
College of Medicine
Department: Anatomy &
Pathology***

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Objectives:

- Histology is the science that deals with the study of the body tissues & how these tissues are arranged to constitute organs.
- Student must understand why we study histology.
- Student must understand how the histological sections are prepared for study by microscope.
- Student must learn what are the basic types of tissues in human body.
- General characteristic of epithelial tissue and major types of epithelial cells.

Histology

is the study of cells, tissues and organs as seen under a microscope.
The purpose of this science is to

- 1-understanding tissue structure at level not visible to the naked eye.
- 2- for understanding the relationship between tissue structure and functions.
- 3-for understanding the differences between normal & abnormal tissue structure which is essential in histopathology.
- 4-providing a basis for the diagnosis and treatment of disease & injured tissue.

Preparation of histological sections

Cutting

Fixation

**Dehydration and
Clearing**

Embedding

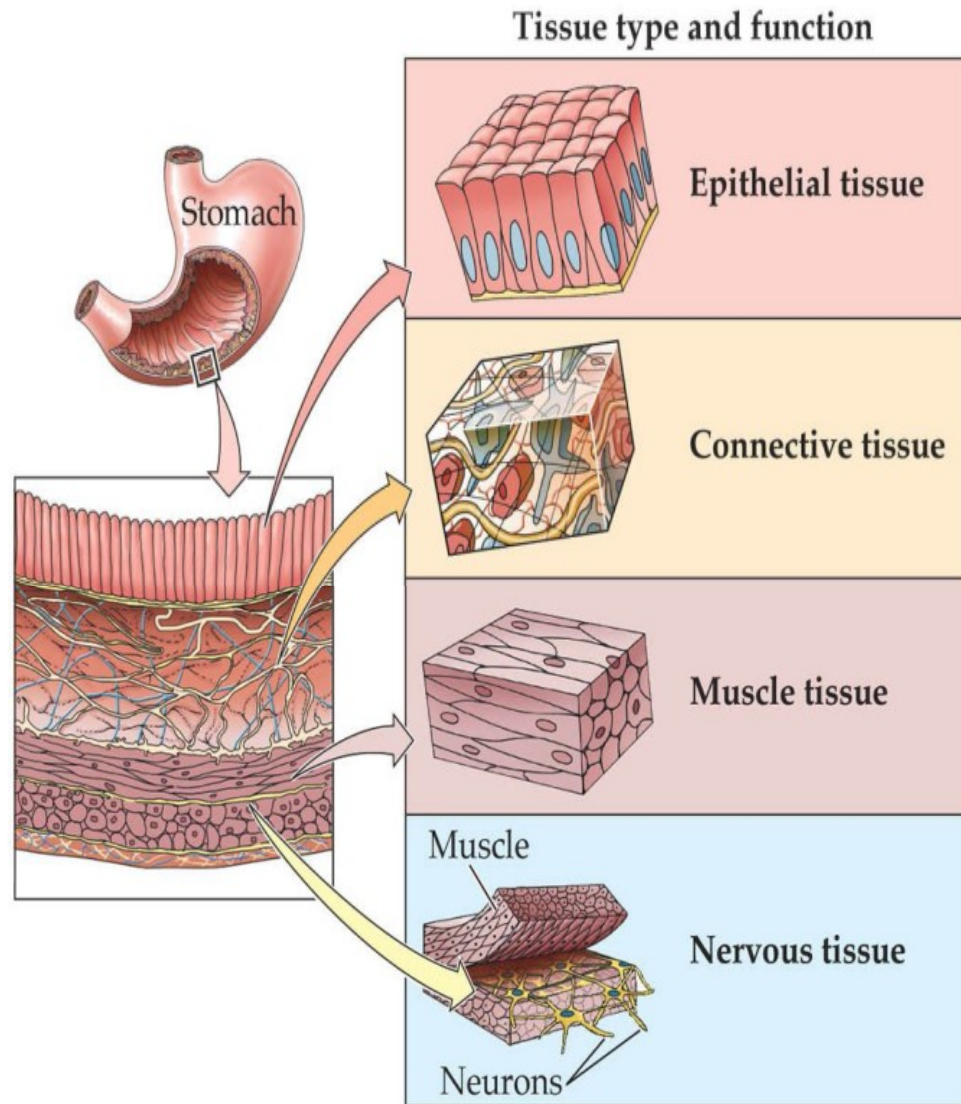
Sectioning

Mounting

Staining

Tissue

- Cells are grouped into tissues and each tissue type is highly specialized to perform specific functions.
- The human body is composed of four basic types of tissues:
 - 1- epithelial tissue
 - 2-connective tissue
 - 3-muscular tissue
 - 4-nervous tissue.
- In association with one another tissues form different organs and organ systems of the body.



1- Epithelial tissue.

Epithelium- is a layer of cells that forms a protective barrier and has specific structural and functional properties.

- **Specific properties of epithelium:**
 1. A free surface exposed to the external environment or lining an internal cavity ,chamber , or passage way.
 2. Attachment to underlying connective tissue by basement membrane.

Specific properties of epithelium (continued):

3. The absence of blood vessels(Avascularity). Because epithelium is avascular, epithelial cells obtain nutrients by diffusion from underlying connective tissues or in some case, from their free surface.
- 4- Cells are tightly packed together,with specialized intercellular junctions and very little Extra Cellular Fluid (ECF).
- 5- High rate of mitosis and regeneration.

Specific properties of epithelium (continued)

6- All epithelial cells are derived from the three embryonic germ layers:

Ectoderm : the epithelium of skin, mouth, nose & anus.

Endoderm: the epithelium of digestive tract & respiratory tract.

Mesoderm: epithelium lining blood vessels (endothelium).

7- Epithelial tissues are continuously renewed due to their regenerative capacity.

Main functions of the epithelial tissues :

1- covering & lining:

- covers external body surfaces (skin).
- lines internal passageways that communicate with the external environment, such as the digestive, respiratory, reproductive and urinary tract.

2-Control permeability, include absorption & secretion.

3- Secretion of specialized substance through *glands*.

4- sensation via (Neuroepithelium)

5- contractility present in specialized epithelial cells eg myoepithelial cells.

6- protection against mechanical injury, dehydration and pathogens.

Major types of epithelial cells:

- 1-Epithelial cells specialized for transport.
- 2-Epithelial cells specialized for absorption.
- 3-Epithelial cells specialized for secretion.
- 4-Epithelial cells specialized for contraction or Myoepithelial cells.

1-Epithelial cells specialized for transport:

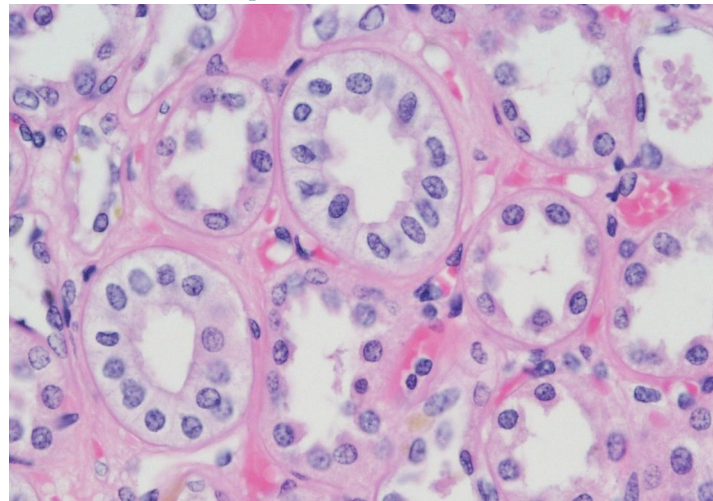
- a. Ion –transporting cells:
- Have highly folded basal plasma membrane that interdigitates with numerous mitochondria reflecting high metabolic activity.
- Found in the kidney tubules, the striated ducts of the salivary glands, the gall bladder, the choroid plexus & the eye's ciliary body.

1-Epithelial cells specialized for transport:

- **b-Cells that transport by pinocytosis:**
- Possess tight junctions and abundant pinocytotic vesicles.
- Transport substances across the cell from luminal(apical) surface to basal surface & vice versa(transcytosis).
- For example the endothelial cells lining the blood vessels

2-Epithelial cells specialized for absorption:

- Possess numerous microvilli on the apical surface which increase the surface area for absorption.
- Commonly found as absorptive cells lining the digestive tracts and the proximal tubules of the kidney.



3-Epithelial cells specialized for secretion

A-Protein secreting cells:

- Contain abundant basophilic RER, a well developed Golgi apparatus, and accumulated secretory granules at the apical region of the cells.

Examples:

- Pancreatic acinar cells, the chief cells of stomach, hepatocytes & chief cells parathyroid glands.

3-Epithelial cells specialized for secretion

- **B-Polypeptide - secreting cells**
- Are smaller than protein secreting cells.
- Show accumulation of secretory granules at the basal region of the cell.
- Belong to APUD system (Amine Precursor Uptake and Decarboxylation cells (APUD))
- Most APUD cells are unicellular glands scattered among other epithelial cells.

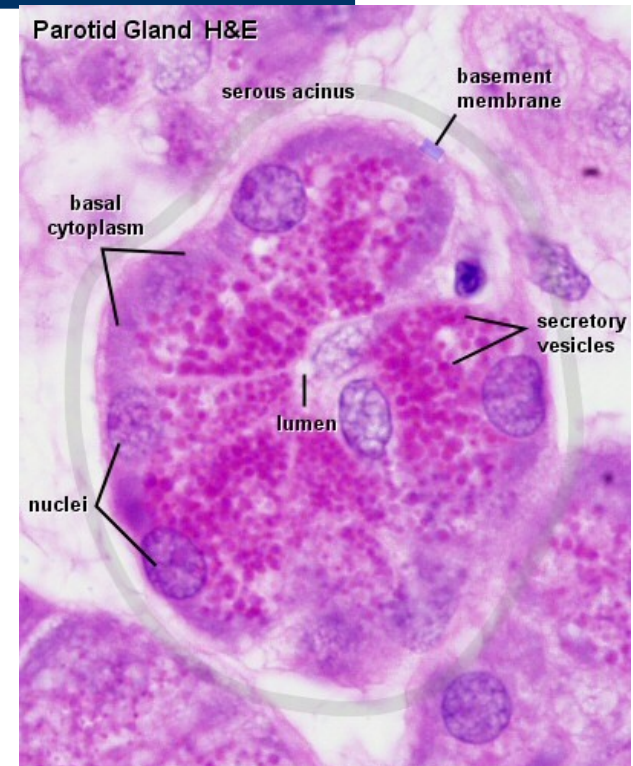
3-Epithelial cells specialized for secretion

- **C-Mucous cells**
- Maybe unicellular (eg, Goblet cells) or arranged in sheets or solid glands.
- Apper light- staining, foamy appearance caused by large mucous containing vesicles concentrated near the apex
- Have weak affinity for H&E staining .
- The nucleus & few (rER) located at the base part of the cell.



3-Epithelial cells specialized for secretion

- **D-Serous cells**
- Share the structural characteristic of protein secreting cells .
- Are smaller, darker staining & more basophilic than mucous- secreting cells.
- Pancreatic acinar cells & the secretory cells of the parotid salivary glands.

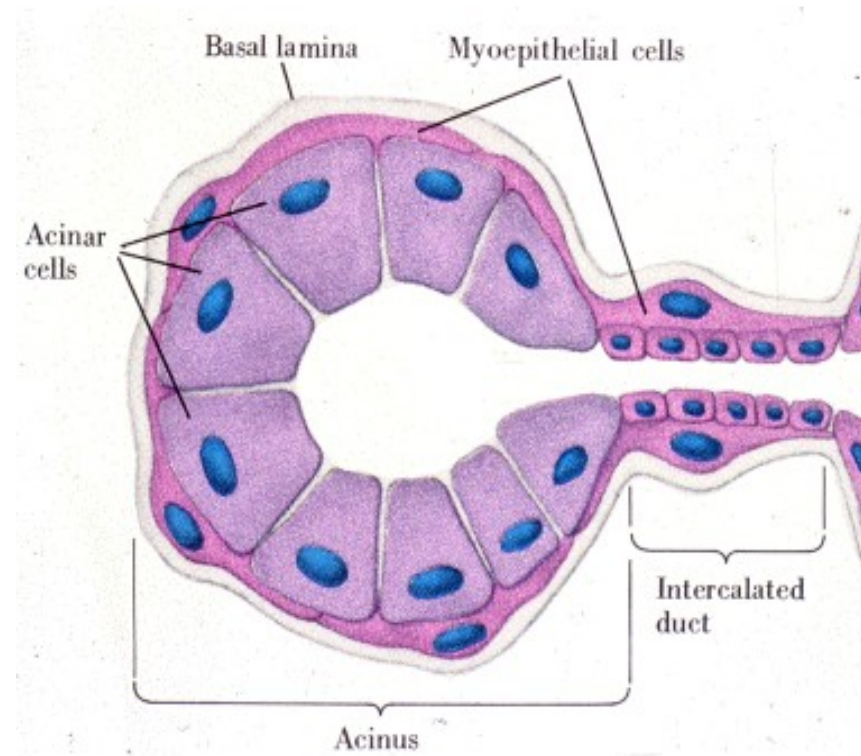


3-Epithelial cells specialized for secretion

- **E-Steroid secreting cells**
- Polygonal or rounded cells with a central nucleus & pale staining acidophilic cytoplasm that contain lipid droplets.
- Testosterone produces by interstitial cells of testes.
- Estrogen produced by ovarian follicular cells (granulosa).
- Progesterone from granulosa lutein cells of the corpus luteum.
- Corticosteroids(eg, Cortisol/cortisone & aldosterone) are produced by cells of adrenal cortex.

4-Contractile epithelial cells or Myoepithelial cells

- Stellate or spindle shaped cells, flattened with finger like cytoplasmic processes that embrace the acinus or duct.
- Locate between the epithelial cells & their basal lamina.
- Found in lacrimal glands & around the seminiferous tubules of the testes, salivary gland and mammary glands.



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Thanks